

Vignettes: Which Test Would You Use?

1 The Executive Function Benchmark Study

Dr. Eleanor Vance, a cognitive psychologist, wants to assess whether the performance of a group of graduate students in her university's demanding neuroscience program differs significantly from the general population on a standardized measure of executive function.

1.1 The Standardized Test

The standardized test for executive function, the Cognitive Flexibility Index (CFI), has been administered to thousands of adults nationwide, establishing clear population parameters:

Population Mean (μ): 100 points

Population Standard Deviation (σ): 15 points

1.2 The Hypothesis and Design

Dr. Vance hypothesizes that the high-achieving, specialized nature of the neuroscience program attracts individuals with above-average executive function.

She administers the CFI to a random sample of 40 graduate students from the program and finds the following results:

Sample Size (N): 40

Sample Mean ($\bar{X}$): 106.5 points

Is the sample mean of 106.5 points significantly higher than the known population mean of 100 points, or is this difference small enough to be attributed to random sampling variation?

2 The Auditory Processing Speed Test

Dr. Chloe Davis, a developmental psychologist, is studying the impact of chronic use of noise-cancelling headphones on auditory processing speed in young adults. She suspects that relying on headphones to filter out environmental noise might slightly reduce the speed at which the brain processes basic sounds, even when the headphones aren't in use.

2.1 The Known Standard

The standard laboratory measure of Auditory Processing Speed (APS) yields a score reflecting the milliseconds required to accurately identify a rapid sequence of tones. Decades of previous research have established that for the general population of young adults:

Hypothesized Population Mean (μ): 45 milliseconds (ms).

The Population Standard Deviation (σ) is difficult to definitively ascertain due to slight variations across labs and testing equipment.

2.2 Design and Hypothesis

Dr. Davis recruits a random sample of 25 young adults who report using noise-cancelling headphones for at least 30 hours per week over the past year.

She administers the APS test to this sample and obtains the following results:

Sample Size (N): 25

Sample Mean ($\bar{X}$): 48.7 ms (indicating slightly slower processing)

Sample Standard Deviation (s): 6.2 ms

Is the mean APS score of 48.7 ms for the headphone group significantly slower than the standard population benchmark of 45 ms, or is this difference just random variation within the sample?

3 The Distraction and Memory Recall Study

Dr. Kenji Tanaka, a cognitive neuroscientist, is investigating the effect of different types of background noise on short-term memory recall. He hypothesizes that auditory distraction (like human speech) impairs memory more severely than visual distraction (like an animated screen saver).

3.1 Design and Hypothesis

Dr. Tanaka recruits 80 volunteer college students and randomly assigns them to one of two conditions:

Group A (Auditory Distraction): $N_A = 40$ participants are asked to memorize a list of 20 words while listening to a recorded conversation.

Group B (Visual Distraction): $N_B = 40$ participants are asked to memorize the same list of 20 words while watching a constantly moving, abstract video.

After a brief interval, both groups complete a memory recall test, where the score is the total number of words correctly recalled (out of 20).

3.2 The Data

Group	Participants (N)	Mean ($\bar{X}$)	Standard Deviation (s)
A (Auditory)	40	12.5 words	3.1 words
B (Visual)	40	14.8 words	2.9 words

Is the 2.3-word difference in mean recall score between the Auditory and Visual distraction groups statistically significant, or is it likely a chance finding?

4 The Introvert-Extrovert Emotional Recognition Study

Dr. Liam Chen, a personality psychologist, is investigating whether introverts and extroverts differ in their ability to accurately recognize subtle, negative emotional expressions. He hypothesizes that due to their different levels of attention to social cues, their performance (and the variability in their performance) might be different.

4.1 Design

Dr. Chen administers a detailed personality survey and recruits two independent groups of students who fall on the extreme ends of the spectrum:

Group A (Introverts): $N_A = 35$ participants

Group B (Extroverts): $N_B = 45$ participants

4.2 The Task and Data

Both groups complete an Emotional Recognition Task (ERT) where they view brief video clips and must score the intensity of a negative emotion (e.g., subtle fear) on a scale of 0 to 100. The score is the average accuracy across all clips.

Group	Participants (N)	Mean ($\bar{X}$)	Standard Deviation (s)	Sample Variance (s^2)
A (Introverts)	35	82.5	14.0	196.0
B (Extroverts)	45	77.1	7.5	56.25

Is the difference in the mean emotional recognition score statistically significant, given that the variability (spread) of scores is substantially different between the two groups?

5 The Mindfulness and Test Anxiety Intervention

Dr. Sarah Nygren, a clinical psychologist, is testing the effectiveness of a brief, four-week mindfulness intervention designed to reduce test anxiety in college students.

5.1 Design

Dr. Nygren recruits a group of 10 students who self-identify as having high levels of test anxiety. She measures their anxiety twice:

1. Measurement 1 (Pre-Intervention): She administers the standardized Test Anxiety Inventory (TAI) to all 10 students before they begin the mindfulness training.
2. Intervention: All 10 students participate in the four-week mindfulness program.
3. Measurement 2 (Post-Intervention): She administers the exact same TAI to the same 10 students immediately after the program concludes.

5.2 Data and Hypothesis

The TAI scores range from 20 (low anxiety) to 80 (high anxiety). The key data are the scores for each individual student before and after the intervention.

Student ID	Pre-Intervention TAI Score	Post-Intervention TAI Score
1	75	68
2	62	60
3	78	70
4	69	65
5	80	72
6	55	50
7	71	67
8	65	59
9	73	64
10	60	58

Did the mindfulness intervention cause a statistically significant reduction in test anxiety? That is, is the difference between the mean Pre-score ($\bar{X}_{\text{Pre}} = 68.8$) and the mean Post-score ($\bar{X}_{\text{Post}} = 63.3$) greater than what would be expected by random chance?

6 Challenge: The Social Conformity Study

Dr. Anya Sharma, a social psychologist, is conducting an experiment to investigate whether subtle visual cues can influence individuals' propensity to conform to a group opinion. She hypothesizes that showing participants a brief, subliminal image of a widely recognized social symbol (like a corporate logo) will increase their likelihood of agreeing with a false consensus.

6.1 The Design

Dr. Sharma recruits 120 college students and randomly assigns them to one of two conditions:

1. Condition A (Subliminal Cue): 60 participants are briefly shown a flash of a popular social media logo before each task.

2. Condition B (Control): 60 participants are briefly shown a flash of a neutral, abstract geometric shape before each task.

6.2 The Task

Participants in both conditions are presented with a series of visual perception tasks (e.g., estimating the length of a line) but are first shown the unanimously incorrect answers of five supposed “prior participants” (who are actually confederates of the experimenter). Dr. Sharma records whether or not the participant conformed by giving the incorrect, majority answer.

6.3 The Data

Condition	Participants (N)	Participants Who Conformed (X)
A (Subliminal Cue)	60	45
B (Control)	60	33

Is the 20% increase in conformity in the subliminal cue group statistically significant, or is it merely due to random chance?